End Matter (draft): A temporally-constrained latent space reveals the stochastic dynamics of convective organization

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(Dated: July 28, 2026)

PER-STATE MASS-FLUX PROFILES

Let $M(h, z, t)$ denote the isentropic mass flux at time t, height z, and moist static energy h. For each cluster c, we define the cluster-mean vertically integrated positive and negative mass fluxes,

$$M_c^{\pm}(z) = \left\langle \int_h M(h, z, t)\, \mathbf{1}\{\pm M > 0\} \right\rangle_{t \in c}, \tag{S1}$$

and the anomalies relative to the simulation-wide mean profiles $\overline{M}^{\pm}(z)$,

$$\Delta_c^{\pm}(z) = M_c^{\pm}(z) - \overline{M}^{\pm}(z). \tag{S2}$$

The vertical profiles (Fig. S1, left) reveal sharp dynamical distinctions. State 1 has a deep upward anomaly with only a weak low-level downdraft, consistent with sustained deep ascent and suppressed detrainment; state 2 shows an even stronger upper-tropospheric ascent with the deepest mid-level downdraft, the signature of a mesoscale-stratiform system; state 4 peaks at lower altitude, characteristic of active but weakly organized convection; state 5 carries only shallow low-level ascent; and state 3 shows weak anomalies throughout.

Resolved along moist static energy (Fig. S1, right), the states separate thermodynamically. State 1 peaks at high $\theta \approx 343$ K (warm, moist air lifted with reduced entrainment); state 2 spreads across $\theta \approx 330$–345 K with its positive peak at intermediate θ (detrainment into a stratiform region); state 4 concentrates at lower $\theta \approx 329$ K (less buoyant air, stronger entrainment); state 5 is confined to low–intermediate θ; and state 3 lies below the mean throughout.

AUTOCORRELATION OF STATE INDICATORS IN A MARKOV CHAIN

Let $\{X_t\}_{t \geq 0}$ be a stationary Markov chain on $\{1, \ldots, K\}$ with transition matrix $\mathbf{T}$ and stationary distribution $\boldsymbol{\pi}$ ($\mathbf{T}\boldsymbol{\pi} = \boldsymbol{\pi}$, $\sum_i \pi_i = 1$). For state i, the indicator $I_i(t) = \mathbb{1}[X_t = i]$ has mean $\mathbb{E}[I_i(t)] = \pi_i$, and we define the normalized autocorrelation

$$C_i(\tau) = \frac{\mathbb{E}[(I_i(t) - \pi_i)(I_i(t+\tau) - \pi_i)]}{\mathbb{E}[(I_i(t) - \pi_i)^2]}. \tag{S3}$$

Since I_i is Bernoulli, the denominator is $\pi_i(1 - \pi_i)$; by stationarity and the Markov property $\mathbb{E}[I_i(t) I_i(t+\tau)] = \pi_i (\mathbf{T}^{\tau})_{ii}$. Hence

$$C_i(\tau) = \frac{(\mathbf{T}^{\tau})_{ii} - \pi_i}{1 - \pi_i}. \tag{S4}$$

Writing the spectral decomposition $\mathbf{T}^{\tau} = \mathbf{1}\boldsymbol{\pi}^{\top} + \sum_{\alpha=2}^{K} \lambda_\alpha^{\tau}\, \mathbf{v}_\alpha \mathbf{w}_\alpha^{\top}$, with eigenvalues $1, \lambda_2, \ldots, \lambda_K$ ($|\lambda_\alpha| < 1$ for $\alpha \geq 2$) and right/left eigenvectors $\mathbf{v}_\alpha, \mathbf{w}_\alpha$, the stationary term cancels and

$$C_i(\tau) = \sum_{\alpha=2}^{K} A_\alpha^{(i)} \lambda_\alpha^{\tau}, \qquad A_\alpha^{(i)} = \frac{(\mathbf{v}_\alpha)_i (\mathbf{w}_\alpha)_i}{1 - \pi_i}. \tag{S5}$$

At large lag the second eigenvalue dominates, $C_i(\tau) \sim \lambda_2^{\tau} = \exp(-\tau/\tau_{\mathrm{corr}})$ with $\tau_{\mathrm{corr}} = -1/\ln|\lambda_2|$: exponential decay of the state autocorrelations is a direct consequence of the Markov property.

COLD-POOL DETECTION AND ACTIVITY INDEX

Cold pools, masses of cold, dense air that form below convective clouds through rain evaporation, are a key control on squall-line organization: they outlast the downdrafts that generate them and trigger or suppress new convection at their gust fronts [1–5]. In the main text they are identified as a candidate carrier of the memory that keeps the organized state 1 (and the active-unorganized state 4) alive beyond the Markov prediction. Testing this requires a time-resolved measure of cold-pool activity: we detect cold pools at each time step and summarize their total extent through a cold-pool index.

Cold-pool detection

Cold pools are detected at each model time step from the two-dimensional surface temperature field $T_{abs}(x, y, t)$, smoothed with a Gaussian filter ($\sigma = 2$–3 grid points). Two thresholds define cold-pool envelopes ($T_{abs} < \phi_{\mathrm{env}}$) and their colder cores ($T_{abs} < \phi_{\mathrm{core}} < \phi_{\mathrm{env}}$). Envelope points are grouped into connected objects by single-linkage clustering with a periodic metric;



FIG. S1. **State-averaged mass-flux anomalies. Left:** vertically resolved anomalies, **(a)** positive (upward) and **(b)** negative (downward). **Right:** the same anomalies resolved along moist static energy θ. All are relative to the simulation-wide mean profile, computed from the samples closest to each state centroid in latent space; colours denote states (legend), solid lines the per-cluster mean, and shaded bands the 90% bootstrap confidence interval.

core labels are propagated outward by nearest-neighbor classification, giving a labeled cold-pool mask at each step.

Cold-pool index

A cold-pool index $\mathrm{CP}(t)$ is defined as the total area covered by cold pools at each time step,

$$\mathrm{CP}(t) = \sum_k A_k(t), \tag{S6}$$

where the sum runs over all cold pools present at time t. Normalized, anomalous, and standardized versions are also computed from its temporal mean and standard deviation.

Cold-pool conditioning of state-1 survival

If cold pools carry the memory that keeps state 1 alive beyond the Markov prediction, then a state-1 configuration that *already* carries a strong cold pool should behave *more* Markovianly: much of the extra persistence is then explained by the cold pool that is present, so the state-1 autocorrelation should decay faster, back toward the Markov reference. We test this by conditioning on the cold-pool index at the starting time.

We first select the state-1 timesteps whose cold-pool index exceeds its 75th percentile $\mathrm{CP}_\star$,

$$J_1^\star(t) = \mathbf{1}[X_t = 1 \text{ and } \mathrm{CP}(t) > \mathrm{CP}_\star], \tag{S7}$$

so that $J_1^\star(t) = 1$ flags a "strong–cold-pool" start in state 1. For these starts we follow the *survival* of state 1: the probability that the system is still in state 1 a lag τ later. The indicator $I_1(t+\tau)$ equals 1 when $X_{t+\tau} = 1$ and 0 otherwise, so its conditional expectation given $J_1^\star(t) = 1$—an average of 0s and 1s over the selected starts—is exactly the probability that the trajectory is back in (or still in) state 1 at lag τ,

$$S_1^{(\mathrm{CP})}(\tau) = \mathbb{E}\big[\, I_1(t+\tau) \,\big|\, J_1^\star(t) = 1 \,\big]. \tag{S8}$$

By construction $S_1^{(\mathrm{CP})}(0) = 1$, since every selected trajectory starts in state 1; and at long lag the memory of the start is lost, so $S_1^{(\mathrm{CP})}(\tau) \to \pi_1$, the stationary occupancy of state 1. To place this survival on the same scale as the unconditional autocorrelation $C_1(\tau)$ of Eq. (S3), which runs from 1 at $\tau = 0$ to 0 at large lag, we subtract the baseline π_1 and divide by the available range $1 - \pi_1$,

$$C_1^{(\mathrm{CP})}(\tau) = \frac{S_1^{(\mathrm{CP})}(\tau) - \pi_1}{1 - \pi_1}. \tag{S9}$$

Subtracting π_1 sends the long-lag limit to 0 and dividing by $1 - \pi_1$ sends $\tau = 0$ to 1, mapping the survival from $[\pi_1, 1]$ onto $[0, 1]$ so it shares the axes of $C_1(\tau)$ and the Markov prediction (Fig. 5b).

The condition is imposed *only at the starting time* t; the trajectory then evolves freely—holding a cold pool in place throughout would force state 1 to persist and artificially truncate the survival, confounding the memory we seek to isolate.

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